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You are an expert lecture scribe generator

You have to generate a complete lecture scribe intended to serve as exam-oriented reference material.

This scribe must allow a student to fully reconstruct the entire lecture for a closed-notes, reading-based exam using this document as the sole reference.

Mandatory context and scope constraints :

Use only the provided lecture slides and permitted textbook material as context.

Treat the provided lecture material as complete, ordered, and mandatory.

Cover the lecture from the FIRST instructional slide to the FINAL instructional slide.

Do NOT stop early.

Do NOT assume logical stopping points.

Every section, subsection, definition, assumption, theorem, proposition, example, derivation, proof, equation, table, figure, diagram, and stated result appearing anywhere in the context MUST be included.

If new concepts appear later in the lecture, they MUST also be included.

MANDATORY CONTENT RULES:

Do NOT summarize, compress, or omit material.

Do NOT skip material even if sections appear conceptually independent.

ALL worked examples must be written fully with intermediate steps exactly as presented.

Proofs and derivations must be written step-by-step exactly as in the lecture.

If a slide contains only statements or definitions, reproduce them formally.

If a slide contains no mathematics, reproduce its content without interpretation.

Do NOT introduce new examples, intuition, motivation, commentary, or explanations.

Do NOT add alternative solution methods.

Do NOT add background knowledge.

Do NOT add transitional or conversational text.

Do NOT include anything not directly traceable to the provided context.

STRUCTURE REQUIREMENT:

Use EXACTLY the same structure as the example lecture scribe below.

Do NOT rename sections.

Do NOT reorder sections.

Do NOT add or remove sections.

Example Lecture Scribe (structure only, do not copy content):

Topic Title

Overview

Brief statement describing what the lecture covers.

Definitions and Notation

Formal definitions and symbols introduced in the lecture.

Main Results

Statements of theorems, propositions, or key results, section-wise. Include figures, graphs, tables, or diagrams ONLY if they explicitly appear in the lecture.

If a figure is referenced in the lecture, include a formal textual description exactly as stated.

Derivations / Proofs

Step-by-step derivations or proofs exactly as presented.

Worked Examples

Example problems solved in class with all intermediate steps.

Summary

Concise recap strictly limited to formulas, results, and definitions already stated.

FINAL INSTRUCTION:

Follow the lecture order strictly from beginning to end.

The output must be complete rather than concise.

This document must be suitable as the sole exam-reference material.



Topic Title

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

(From Lecture 18 – CSE400, March 12, 2026)

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Overview

This lecture covers:

- Marginal PMF (definition, computation from joint PMF, example)
- Correlation (raw product moment and its limitations)
- Covariance (definition, interpretation, algebraic form, properties, examples)
- Pearson's correlation coefficient
- Joint Gaussian random variables (definition, algebraic form, interpretation, example)

Definitions and Notation

Joint PMF

For discrete random variables X and Y :

$$p_{XY}(x, y) = P(X = x, Y = y)$$

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Marginal PMFs

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

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Correlation (Raw Product Moment)

$$R_{XY} = E[XY]$$

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Covariance

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Equivalent algebraic form:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

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Pearson's Correlation Coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$-1 \leq \rho_{XY} \leq 1$$

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Joint Gaussian Distribution

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

Where:

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : correlation coefficient

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Main Results

Marginal PMF

Key Idea

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.  L18-1-27

From Joint PMF Table

- Row sums give $p_X(x)$
- Column sums give $p_Y(y)$

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

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Correlation

- Measures average value of product XY
- Depends on scale of X and Y
- Computed about origin (not means)
- Mixes:
 - mean values
 - spread
 - dependence

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

Limitation: Not a clean measure of dependence $\rightarrow$ leads to covariance.  L18-1-27

Covariance

Interpretation

- Measures joint variation after centering
- Positive covariance: deviations in same direction
- Negative covariance: opposite directions
- Zero covariance: no linear co-variation

Magnitude depends on units $\rightarrow$ cannot measure strength.  L18-1-27

Algebraic Result

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Properties

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as special case

$$\text{Cov}(X, X) = \text{Var}(X)$$

3. Linearity

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

4. Independence implies zero covariance

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

Note: Zero covariance does **not** imply independence.  L18-1-27

Pearson's Coefficient

- Measures **strength and direction** of linear relationship
- Normalized covariance
- Unit-free

Interpretation:

- $\rho > 0$: positive linear trend
- $\rho < 0$: negative linear trend
- $\rho = 0$: no linear correlation

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Joint Gaussian RV

Definition:

A pair (X, Y) is jointly Gaussian if:

- joint density is bivariate Gaussian
- marginals are Gaussian

Applications:

- Signal processing
- Communication systems
- Continuous noisy measurements

Interpretation (example):

- X : CO₂ concentration
- Y : temperature anomaly
- $\rho > 0$: higher CO₂ associated with higher temperature

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Derivations / Proofs

Covariance Derivation

$$\begin{aligned} \text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - X\mu_Y - \mu_X Y + \mu_X \mu_Y] \\ &= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

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Worked Examples

Example 1 (Marginal PMF)

Given joint PMF:

	$y = 0$	$y = 1$
$x = 0$	$1/8$	$1/4$
$x = 1$	$3/8$	$1/4$

Compute marginals

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Thus:

$$p_X(x) = \begin{cases} 3/8 & x = 0 \\ 5/8 & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

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Example 1 (Covariance)

Given:

$$Y = -2X + 3$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$XY = X(-2X + 3) = -2X^2 + 3X$$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Substitute:

$$\text{Cov}(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$

$$= -2E[X^2] + 2(E[X])^2$$

$$= -2(E[X^2] - (E[X])^2)$$

$$= -2\text{Var}(X)$$

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Example 2

Given:

$$f_{XY}(x, y) = \frac{xy}{96}, \quad 0 < x < 4, \quad 1 < y < 5$$

1. Compute $E[X]$

$$\begin{aligned}
 E[X] &= \int_0^4 x f_X(x) dx = \int_0^4 x \cdot \frac{x}{8} dx = \frac{1}{8} \int_0^4 x^2 dx \\
 &= \frac{1}{8} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{8} \cdot \frac{64}{3} = \frac{8}{3}
 \end{aligned}$$

2. Compute $E[Y]$

$$\begin{aligned}
 E[Y] &= \int_1^5 y f_Y(y) dy = \int_1^5 y \cdot \frac{y}{12} dy = \frac{1}{12} \int_1^5 y^2 dy \\
 &= \frac{1}{12} \left[\frac{y^3}{3} \right]_1^5 = \frac{1}{12} \cdot \frac{124}{3} = \frac{31}{9}
 \end{aligned}$$

3. Compute $E[XY]$

$$E[XY] = E[X]E[Y] = \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

4. Compute $E[2X + 3Y]$

$$\begin{aligned}
 E[2X + 3Y] &= 2E[X] + 3E[Y] \\
 &= 2 \cdot \frac{8}{3} + 3 \cdot \frac{31}{9} = \frac{16}{3} + \frac{31}{3} = \frac{47}{3}
 \end{aligned}$$

5. Compute $Var(X)$

$$\begin{aligned}
 E[X^2] &= \int_0^4 x^2 \cdot \frac{x}{8} dx = \frac{1}{8} \int_0^4 x^3 dx \\
 &= \frac{1}{8} \cdot \left[\frac{x^4}{4} \right]_0^4 = \frac{1}{8} \cdot 64 = 8 \\
 Var(X) &= 8 - \left(\frac{8}{3} \right)^2 = 8 - \frac{64}{9} = \frac{8}{9}
 \end{aligned}$$

6. Compute $Var(Y)$

$$\begin{aligned}
 E[Y^2] &= \frac{1}{12} \int_1^5 y^3 dy = \frac{1}{12} \cdot \left[\frac{y^4}{4} \right]_1^5 \\
 &= \frac{1}{12} \cdot \frac{624}{4} = 13 \\
 Var(Y) &= 13 - \left(\frac{31}{9} \right)^2 = 13 - \frac{961}{81} = \frac{92}{81}
 \end{aligned}$$

7. Compute Covariance

$$Cov(X, Y) = \frac{248}{27} - \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27} - \frac{248}{27} = 0$$

8. Compute Correlation

$$\rho_{XY} = \frac{0}{\sqrt{(8/9)(92/81)}} = 0$$

Summary

- Marginal PMF:

$$p_X(x) = \sum_y p_{XY}(x, y), \quad p_Y(y) = \sum_x p_{XY}(x, y)$$

- Correlation (raw):

$$R_{XY} = E[XY]$$

- Covariance:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- Properties: symmetry, variance special case, linearity, independence $\Rightarrow$ zero covariance
- Pearson coefficient:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Joint Gaussian PDF defined with parameters $\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho$

Sources result: